



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta 27/01/2022*

PROBLEM - We observe N samples of the following discrete-time real random process:

$$X[n] = S[n] + W[n] = A \cos(2\pi F_0 n - \theta) + W[n],$$

where A and θ are mutually independent random parameters and $F_0 \in (0, 1/2)$ is a-priori known, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) $S_W(e^{j2\pi f}) = \sigma_W^2$, where σ_W^2 is a-priori known. A is a Rayleigh distributed random variable with power $E\{A^2\} = \sigma_A^2$, i.e. $f_A(a) = \frac{2a}{\sigma_A^2} e^{-\frac{a^2}{\sigma_A^2}} u(a)$, and θ is uniformly distributed in $[-\pi, \pi)$. A and θ are independent from the noise.

(1) Derive the Signal to Noise power Ratio (SNR) $\gamma \triangleq \frac{E\{S^2[n]\}}{E\{W^2[n]\}}$, the mean value and the autocorrelation function (ACF) of the process $X[n]$.

(2) Derive the joint probability density function (pdf) of the received data vector $\mathbf{X} \triangleq [X[0] \ X[1] \ \dots \ X[N-1]]^T$.

(3) Derive the linear minimum mean square error (LMMSE) estimator of the signal of interest $S[n]$ given the observation of two samples of the process $X[n]$ at times n and $n-1$, i.e. $\hat{S}[n] = h[0]X[n] + h[1]X[n-1]$, and express it as a function of γ and F_0 . Calculate numerically $h[0]$ and $h[1]$ under the assumption that $\gamma = 2$ and $F_0 = 1/4$, and comment on the result.

(4) Derive the Mean Square Error (MSE) $\sigma_\varepsilon^2 \triangleq E\left\{\left(S[n] - \hat{S}[n]\right)^2\right\}$ as a function of σ_A^2 , γ , and F_0 .

Calculate numerically the MSE under the assumption that $\sigma_A^2 = 2$, $\gamma = 2$, and $F_0 = 1/4$.

(5) Derive the *a-posteriori* probability density function (pdf) of $S[n]$ for the parameter values as before: $\sigma_A^2 = 2$, $\gamma = 2$, and $F_0 = 1/4$.



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Solution

(1) The mean value and the ACF are:

$$\eta_x[n] = E\{X[n]\} = 0$$

$$R_x[m] = E\{X[n]X[n+m]\} = R_s[m] + R_w[m] = \frac{\sigma_A^2}{2} \cos[2\pi F_0 m] + \sigma_w^2 \delta[m]$$

Hence, the SNR is given by: $\gamma \triangleq \frac{E\{S^2[n]\}}{E\{W^2[n]\}} = \frac{\sigma_A^2}{2\sigma_w^2}$

(2) The pdf can be derived by observing that $X[n]$ can be expressed as follows:

$$\begin{aligned} X[n] &= A \cos(2\pi F_0 n - \theta) + W[n] = A \cos(\theta) \cos(2\pi F_0 n) + A \sin(\theta) \sin(2\pi F_0 n) + W[n] \\ &= X_1 \cos(2\pi F_0 n) + X_2 \sin(2\pi F_0 n) + W[n] \end{aligned}$$

It can be proved that X_1 and X_2 are Gaussian distributed random variables, so also $X[n]$ is a Gaussian random variable, and as consequence, $\mathbf{X}$ is a Gaussian random vector: $\mathbf{X} \in N(\boldsymbol{\eta}_x, \mathbf{C}_x)$, where:

$$\boldsymbol{\eta}_x = E\{\mathbf{X}\} = \mathbf{0}, \quad \mathbf{C}_x = \mathbf{C}_s + \mathbf{C}_w = \mathbf{C}_s + \sigma_w^2 \mathbf{I}, \quad \text{where } [\mathbf{C}_s]_{i,j} = R_s[j-i] = \frac{\sigma_A^2}{2} \cos[2\pi F_0(j-i)].$$

(3) To find LMMSE estimator we have to apply the Orthogonality Principle: the estimation error $\varepsilon[n] = S[n] - \hat{S}[n]$ should be orthogonal to the data $X[n]$ and also $X[n-1]$.

$$\min_{\{h[n]\}} \sigma_\varepsilon^2 = E\{\varepsilon^2[n]\} \Rightarrow E\{\varepsilon[n]X[n-m]\} = 0, \quad m = 0, 1$$

$$\begin{aligned} E\{\varepsilon[n]X[n-m]\} &= E\left\{\left(S[n] - \hat{S}[n]\right)X[n-m]\right\} = E\left\{\left(S[n] - \sum_{k=0}^1 h[k]X[n-k]\right)X[n-m]\right\} \\ &= E\{S[n]X[n-m]\} - \sum_{k=0}^1 h[k]E\{X[n-k]X[n-m]\} \\ &= R_{sx}[-m] - \sum_{k=0}^1 h[k]R_x[m-k] = 0, \quad m = 0, 1 \end{aligned}$$

Taking into account that in our case $R_{sx}[-m] = R_s[m]$, we finally came up with the linear system of equations:



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$$\sum_{k=0}^1 h[k]R_x[m-k] = R_s[m], \quad m = 0,1$$

that can be expressed in matrix format:

$$\begin{bmatrix} R_x[0] & R_x[1] \\ R_x[1] & R_x[0] \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \end{bmatrix} = \begin{bmatrix} R_s[0] \\ R_s[1] \end{bmatrix}$$

which in our case becomes:

$$\begin{bmatrix} \frac{\sigma_A^2}{2} + \sigma_w^2 & \frac{\sigma_A^2}{2} \cos[2\pi F_0] \\ \frac{\sigma_A^2}{2} \cos[2\pi F_0] & \frac{\sigma_A^2}{2} + \sigma_w^2 \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \end{bmatrix} = \begin{bmatrix} \frac{\sigma_A^2}{2} \\ \frac{\sigma_A^2}{2} \cos[2\pi F_0] \end{bmatrix}$$

$$\frac{\sigma_A^2}{2} \begin{bmatrix} 1+1/\gamma & \cos[2\pi F_0] \\ \cos[2\pi F_0] & 1+1/\gamma \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \end{bmatrix} = \frac{\sigma_A^2}{2} \begin{bmatrix} 1 \\ \cos[2\pi F_0] \end{bmatrix}$$

$$\begin{bmatrix} h[0] \\ h[1] \end{bmatrix} = \begin{bmatrix} 1+1/\gamma & \cos[2\pi F_0] \\ \cos[2\pi F_0] & 1+1/\gamma \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ \cos[2\pi F_0] \end{bmatrix}$$

$$\begin{bmatrix} h[0] \\ h[1] \end{bmatrix} = \frac{1}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]} \begin{bmatrix} 1+1/\gamma & -\cos[2\pi F_0] \\ -\cos[2\pi F_0] & 1+1/\gamma \end{bmatrix} \cdot \begin{bmatrix} 1 \\ \cos[2\pi F_0] \end{bmatrix}$$

$$\begin{cases} h[0] = \frac{(1+1/\gamma) - \cos^2[2\pi F_0]}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]} \\ h[1] = \frac{-\cos[2\pi F_0] + (1+1/\gamma)\cos[2\pi F_0]}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]} = \frac{1/\gamma \cos[2\pi F_0]}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]} \end{cases}$$



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$$\gamma = 2, F_0 = 1/4 \Rightarrow \cos\left[2\pi\frac{1}{4}\right] = 0 \Rightarrow \begin{cases} h[0] = \frac{(1+1/\gamma)}{(1+1/\gamma)^2} = \frac{1}{1+1/\gamma} = \frac{\gamma}{\gamma+1} = \frac{2}{3} = 0.667 \\ h[1] = 0 \end{cases}$$

(4) The MSE is obtained by applying again the Orthogonality Principle:

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{\varepsilon[n]S[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} \\ &= E\{S^2[n]\} - \sum_{k=0}^1 h[k]E\{S[n]X[n-k]\} \\ &= R_s[0] - \sum_{k=0}^1 h[k]R_s[k] \\ &= \frac{\sigma_A^2}{2}(1 - h[0] - h[1]\cos[2\pi F_0]) \end{aligned}$$

Assumption: $\sigma_A^2 = 2$, $\gamma = 2$, and $F_0 = 1/4$

$$\Rightarrow \sigma_\varepsilon^2 = \frac{\sigma_A^2}{2}(1 - h[0] - h[1]\cos[2\pi F_0]) = 1 - h[0] = \frac{1}{3} = 0.3333$$

(5) Since we are in the Gaussian case (the random parameter to be estimated and the data vector are jointly Gaussian), the *a-posteriori* probability density function (pdf) of $S[n]$ is given by:

$$S[n]|\mathbf{X} \in N(\hat{S}[n], \sigma_\varepsilon^2) \equiv N(0.667 \cdot X[n], 0.3333)$$